AQ3.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Choose the set of correct options.

- [ ] The set $\{(1,2,3),(4,5,6),(7,8,9)\}$ is linearly dependent.
- [ ] The set $\{(1,2,3),(4,5,6),(5,7,9)\}$ is linearly independent.
- [ ] The set $\{(1,2,3),(0,5,6),(0,0,9)\}$ is linearly independent.
- [ ] The set $\{(1,2,3),(0,0,6),(0,0,9)\}$ is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The set $\{(1,2,3),(4,5,6),(7,8,9)\}$ is linearly dependent.
The set $\{(1,2,3),(0,5,6),(0,0,9)\}$ is linearly independent.

1 point
If $ad - bc = 0$ then which of the following options are true?

- [ ] The set $\{(a,b),(c,d)\}$ is linearly dependent in R^2.
- [ ] The set $\{(a,c),(b,d)\}$ is linearly dependent in R^2.
- [ ] The set $\{(1,0,0),(0,a,b),(0,c,d)\}$ is linearly dependent in R^3.
- [ ] The set $\{(1,0,0),(0,a,c),(0,b,d)\}$ is linearly dependent in R^3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The set $\{(a,b),(c,d)\}$ is linearly dependent in R^2.
The set $\{(a,c),(b,d)\}$ is linearly dependent in R^2.
The set $\{(1,0,0),(0,a,b),(0,c,d)\}$ is linearly dependent in R^3.
The set $\{(1,0,0),(0,a,c),(0,b,d)\}$ is linearly dependent in R^3.

For $c \in Z$, consider the set of three vectors $S = \{(1,c,0),(c,0,c),(2c,3c,4c)\}$ in R^3 with usual addition and scalar multiplication. Find the value of c such that the given set is linearly dependent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point
1 point

Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly independent and consider the following system of linear equations

$$\begin{aligned} a_1 x + b_1 y + c_1 z &= 1 \\ a_2 x + b_2 y + c_2 z &= 2 \\ a_3 x + b_2 y + c_3 z &= 3. \end{aligned}$$

z = 3.

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then which of the following option(s) is(are) true?

- [ ] The system of linear equations has a unique solution.
- [ ] The system of linear equations has no solution.
- [ ] The system of linear equations have infinitely many solutions.
- [ ] The coefficient matrix A is invertible.
- [ ] The coefficient matrix A is not invertible.
- [ ] Any one of the above options cannot be concluded using the given information.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The system of linear equations has a unique solution.
The coefficient matrix A is invertible.
1 point

Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly dependent and consider the following system of linear equations

$$\begin{aligned} a_1 x + b_1 y + c_1 z &= 1 \\ a_2 x + b_2 y + c_2 z &= 2 \\ a_3 x + b_2 y + c_3 z &= 3. \end{aligned}$$

z = 3.

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then which of the following option(s) is(are) true?

- [ ] Either the system of linear equations has a unique solution or no solution.
- [ ] Either the system of linear equations has no solution or infinitely many solutions.
- [ ] Either the system of linear equations have infinitely many solutions or a unique solution.
- [ ] The system of linear equations has a unique solution.
- [ ] The coefficient matrix A is invertible.
- [ ] The coefficient matrix A is not invertible.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Either the system of linear equations has no solution or infinitely many solutions.
The coefficient matrix A is not invertible.
If S is a linearly independent set in vector space R^3 with respect to usual addition and scalar multiplication, then find the maximum possible cardinality of S.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Level 2

1 point
Let the set $\{(a, b), (c, d)\}$ be a linearly independent subset of R^2. Choose the set of correct options.

- [ ] $\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of R^3.
- [ ] $\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of R^3.
- [ ] $\{(a, b, 0), (c, d, 0), (1, 0, 0)\}$ must be a linearly independent subset of R^3.
- [ ] $\{(a, b, 0), (c, d, 0), (0, 1, 0)\}$ must be a linearly independent subset of R^3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of R^3.
$\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of R^3.
1 point
Each column of an $n \times n$ matrix A can be thought of as vectors in R^n. If a set S consists of all the column vectors of A, then which of the following is(are) true?

- [ ] S is always linearly independent.
- [ ] S is linearly independent if $det(A) \neq 0$.
- [ ] S is linearly independent if $det(A) = 0$.
- [ ] If $det(A) = 1$, then S is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
S is linearly independent if $det(A) \neq 0$.
If $det(A) = 1$, then S is linearly independent.
1 point

Suppose it is known that the three vectors $v_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, v_2 = \begin{bmatrix} d \\ e \\ f \end{bmatrix}, v_3 = \begin{bmatrix} g \\ h \\ i \end{bmatrix}$ are linearly dependent in R^3. Which of the following is correct?

- [ ] Some linear combination of a, b, c must be 0.
- [ ] Some linear combination of a, d, g must be 0.
- [ ] Some linear combination of d, e, f must be 0.
- [ ] Some linear combination of c, f, i must be 0.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Some linear combination of a, b, c must be 0.
Some linear combination of a, d, g must be 0.
Some linear combination of d, e, f must be 0.
Some linear combination of c, f, i must be 0.

If vectors $(2x, 4, -6), (-6, -3x, 2x), (-1, x-3, x+2)$ from the vector space R^3 with usual addition and scalar multiplication, are linearly dependent and $x \in [2, 5]$, then find the value of x.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 2

1 point